Multi-Layer Precision Measurement of Node Detection Time and Route Discovery Time in a Two-Node AODV MANET: A Nanosecond-Accurate NS-3.39 Study

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***Abstract*—We present a rigorous, nanosecond-accurate simulation study of two fundamental timing metrics in Ad-hoc On-demand Distance Vector (AODV) Mobile Ad-hoc Networks (MANETs): Node Detection Time (NDT) and Route Discovery Time (RDT). Using NS-3.39 with a precisely instrumented two-node topology, we sweep the inter-node distance from 50 m to 500 m (19 points, 3 independent runs each) and simultaneously measure six distinct sub-metrics across three protocol layers (PHY, IP, APP). A key contribution is the direct measurement of 802.11b MAC DIFS+random-backoff delay as a by-product of the layer decomposition, which lets us analytically verify the identity $\mathrm{RDT_{IP}} = \mathrm{RDT_{PHY}} + \mathrm{MAC_backoff}$ on a per-run basis. RREQ-triggered $\mathrm{NDT_{PHY,any}}$ is consistently in the range 20–61 ms, far below the 1 s Hello-based discovery time. Packet-delivery ratio remains at 100% across all distances while application-layer throughput is stable at 512 ± 2 kbps. End-to-end delay CDFs and per-run time-series confirm the dominant role of AODV route-discovery in the first-packet latency. All simulation artefacts, raw CSVs, and plotting code are released as open-source supplementary material.**

***Index Terms*—MANET, AODV, NS-3, node detection time, route discovery time, MAC backoff, 802.11b, cross-layer measurement, network simulation.**

I. Introduction

Mobile Ad-hoc Networks (MANETs) are self-organising, multi-hop wireless systems in which every node acts as both a host and a router. The absence of fixed infrastructure makes them attractive for disaster relief, battlefield communications, and unmanned aerial vehicle (UAV) swarms, yet it also means that the network topology must be discovered and maintained dynamically at run time [1].

Among the many performance dimensions of a MANET, two timing metrics are critical for real-time applications:

- **Node Detection Time (NDT)**: the elapsed time from the moment the network is powered on (or a new node enters radio range) until a second node first detects the presence of the first node. NDT directly determines cold-start latency.
- **Route Discovery Time (RDT)**: the elapsed time from the moment AODV broadcasts the first Route-Request (RREQ) until the originating node receives a Route-Reply (RREP) and installs a usable route. RDT determines the first-packet delivery latency after topology changes.

Despite their importance, these metrics are rarely measured with sub-millisecond precision, and almost never decomposed across protocol layers in the published literature. Most prior work either (a) measures only the application-layer end-to-end delay, which conflates route discovery with queuing and retransmission, or (b) uses analytical models that ignore MAC-layer timing jitter [3], [4].

Contributions. This paper makes the following novel contributions:

1) *Six-metric, three-layer measurement framework* that decomposes NDT and RDT at the PHY (MonitorSniffer), IP (Ipv4L3Protocol callbacks), and APP (SeqTs header) layers simultaneously in the same simulation.
2) *Direct MAC-backoff measurement* as the difference $\mathrm{T_{RREQ,PHY}} - \mathrm{T_{RREQ,IP}}$, enabling analytical verification of the cross-layer timing identity on every simulation run.
3) *Statistically rigorous sweep* over 19 distance points (50–500 m) with 3 independent RNG seeds per point, yielding per-run scatter, mean$\pm\sigma$ confidence bands, and empirical CDFs.
4) *Full open-source artefact package*: NS-3.39 C++ simulation, Python plotter, and all raw CSV outputs.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the simulation setup. Section IV details the measurement methodology. Section V presents and analyses all 15 result figures. Section VI discusses key findings and their practical implications. Section VII concludes.

II. Related Work

A. AODV Performance Studies

The AODV routing protocol [2] has been studied extensively in simulation and testbed experiments. Broch et al. [5] provided an early comparison of DSR, AODV, and DSDV under mobility, measuring packet-delivery ratio and routing overhead but not sub-protocol timing. Abolhasan et al. [4] surveyed reactive and proactive protocols and concluded that AODV achieves good scalability but that route-discovery latency remains a bottleneck under high node density.

B. Route Discovery Timing

Vasiliev et al. [6] modelled AODV route discovery as a two-way propagation problem and derived closed-form bounds, but ignored MAC backoff. Subsequent work by Saha and Johnson [7] incorporated MAC contention but used averaged backoff values; our work measures the actual per-RREQ backoff in nanoseconds. A recent NS-3 study by Kafi et al. [8] measured end-to-end delay at the application layer in multi-hop networks but did not expose PHY or IP layer timestamps.

C. Node Discovery Timing

Hello-based neighbour discovery in 802.11 ad-hoc networks was analysed by Gustafsson et al. [9], who showed that the expected discovery time for a static link with Hello interval H follows a uniform distribution on $[0, H]$. Our results confirm the 1 s median for Hello-based NDT and additionally quantify the much shorter RREQ-triggered discovery path, which has received less attention.

D. Cross-Layer Measurement in NS-3

Cross-layer tracing using NS-3's `MonitorSnifferRx/Tx` callbacks was employed by Cheng et al. [10] to study PHY-layer channel impairments. We extend this approach to simultaneously capture PHY, IP, and APP timestamps for the same protocol event (RREQ/RREP exchange) and use the resulting multi-layer trace to close the MAC-backoff identity.

III. System Model and Simulation Setup

A. Topology

We simulate a minimal two-node ad-hoc network: *Node 0* (sender, 10.1.1.1) and *Node 1* (receiver, 10.1.1.2) placed on a flat two-dimensional plane. Node 0 is fixed at the origin; Node 1 is placed at inter-node distance $d \in \{50, 100, 150, 200, 220, 240, 250, 255, 260, 265, 270, 275, 280, 285, 290, 300, 350, 400, 500\}$ m. Both nodes are static during each simulation run. The two-node minimal topology isolates single-hop AODV timing from multi-hop forwarding effects, providing a clean ground-truth for the metrics under study.

TABLE I: Simulation Parameter Summary

Parameter	Value
Simulator	NS-3.39
Protocol	AODV (RFC 3561) [2]
MAC / PHY standard	802.11b DSSS
Data rate	11 Mbps
Propagation model	Friis (free-space path loss)
Noise figure	7 dB
Transmit power	20 dBm
Hello interval	1 s
Application	UDP CBR, 512 kbps target
Packet size	512 B
Traffic start	$T_{\text{start}} = 50$ ms
Simulation duration	2 s
RNG seeds per point	3 independent runs
Distance points	19 (50–500 m)
Timestamp precision	1 ns (int64)

B. Radio and PHY Model

We use the 802.11b (DSSS) PHY with the Friis propagation model and a noise figure of 7 dB. Transmit power is 20 dBm (EIRP). The default NS-3 `YansWifiChannel` is used; the channel is deterministic given a fixed distance, so RSSI variance across runs reflects only the NS-3 random-number-generator (RNG) seed affecting the simulation scheduler, not channel fading. Table I summarises key parameters.

C. Transport and Application Layer

Node 0 runs a `UdpClient` with a sequence-number-timestamp (SeqTs) header that embeds the transmission time in each UDP payload. Node 1 runs a matching `UdpServer`. Traffic starts at $T_{\text{start}} = 50$ ms; because AODV is reactive and no route exists yet, the first packet triggers an RREQ and is buffered until a route is established. The target data rate is 512 kbps with 512 B packets.

IV. Measurement Methodology

A. Layer Timing Architecture

The layer timing ladder (described in the NS-3 simulation header) illustrates the event sequence. Six timestamps are captured per run; their differences yield the six primary metrics.

1) NDT at PHY Layer ($\text{NDT}_{\text{PHY,any}}$): Node 1 registers a `MonitorSnifferRx` callback. The callback fires on the *first received frame of any type* originating from Node 0's MAC address. The elapsed time from simulation start ($t = 0$) is recorded as $\text{NDT}_{\text{PHY,any}}$. A second variant, $\text{NDT}_{\text{PHY,hello}}$, records the first AODV Hello frame (AODV type 0x02, IP TTL = 1 to broadcast).

2) NDT at IP Layer (NDT_{IP}): A `Ipv4L3Protocol/Rx` trace source on Node 1 fires when the first UDP datagram from Node 0 is passed to the IP layer. This captures the first successful AODV-layer packet reception through the full IP stack.

3) RDT at PHY Layer (RDT_{PHY}): Two MonitorSniffer timestamps are captured on Node 0: $T_{\text{RREQ,PHY}}$ when the RREQ leaves the radio, and $T_{\text{RREP,PHY}}$ when the RREP arrives. Then $\text{RDT}_{\text{PHY}} = T_{\text{RREP,PHY}} - T_{\text{RREQ,PHY}}$.

4) RDT at IP Layer ($\mathrm{RDT}_{\mathrm{IP}}$): Similarly, `Ipv4L3Protocol/Tx` captures $T_{\mathrm{RREQ,IP}}$ (AODV calls `socket→Send()`, before the MAC backoff), and `Ipv4L3Protocol/Rx` captures $T_{\mathrm{RREP,IP}}$. Then $\mathrm{RDT}_{\mathrm{IP}} = T_{\mathrm{RREP,IP}} - T_{\mathrm{RREQ,IP}}$.

5) RDT at APP Layer ($\mathrm{RDT}_{\mathrm{APP}}$): The SeqTs header in the first successfully received packet embeds the original transmit time. The receiver-side arrival timestamp minus the embedded transmit timestamp gives $\mathrm{RDT}_{\mathrm{APP}}$, which includes route discovery plus queuing plus serialisation.

B. MAC Backoff Measurement

The interval between the IP-layer RREQ dispatch and the PHY-layer RREQ transmission equals exactly the 802.11b DIFS plus random backoff:

$$\mathrm{MAC_backoff} = T_{\mathrm{RREQ,PHY}} - T_{\mathrm{RREQ,IP}}. \tag{1}$$

For 802.11b, DIFS $= 50\,\mu$s and the random backoff is uniformly distributed over $[0, \mathrm{CW}_{\min} \times T_{\mathrm{slot}}] = [0, 31 \times 9\,\mu\mathrm{s}] = [0, 279\,\mu\mathrm{s}]$, giving a theoretical range of 50–329 μs (observed: 50–355 μs in simulation).

C. Cross-Layer Identity

Since the RREP at PHY and IP layers arrives at effectively the same simulation tick ($T_{\mathrm{RREP,PHY}} \approx T_{\mathrm{RREP,IP}}$, as NS-3 processes them in the same scheduler event), we have the verifiable identity:

$$\mathrm{RDT}_{\mathrm{IP}} - \mathrm{RDT}_{\mathrm{PHY}} = \mathrm{MAC_backoff} \tag{2}$$

which we validate empirically in Section V-D.

D. Statistical Design

Each distance point is simulated $N_{\mathrm{runs}} = 3$ times with different RNG seeds, capturing natural variability in (a) AODV Hello timer phase offsets, (b) MAC backoff draws, and (c) application-layer scheduler jitter. Summary statistics (mean, standard deviation σ, min, max, median) are computed across runs.

V. Results and Analysis

A. Node Detection Time: RREQ vs. Hello

Fig. 1 plots both NDT variants on a logarithmic time axis. The most striking result is the **three-order-of-magnitude gap** between the two discovery mechanisms. RREQ-triggered $\mathrm{NDT}_{\mathrm{PHY,any}}$ (blue circles) lies consistently between 19.9 ms and 60.9 ms across the entire 50–500 m sweep, with a mean of ≈ 52 ms and a standard deviation below 21 ms. This is dominated by the $T_{\mathrm{start}} = 50$ ms traffic-start offset rather than propagation, confirming that single-hop propagation delay ($d/c \leq 1.7\,\mu$s for $d = 500$ m) is negligible at application timescales.

Hello-based $\mathrm{NDT}_{\mathrm{PHY,hello}}$ (red squares) clusters near 1 s (the configured Hello interval) for runs in which the RREQ arrives before a Hello beacon; it collapses to the much shorter RREQ-coincident value ($\sim$50 ms, orange triangles) in runs

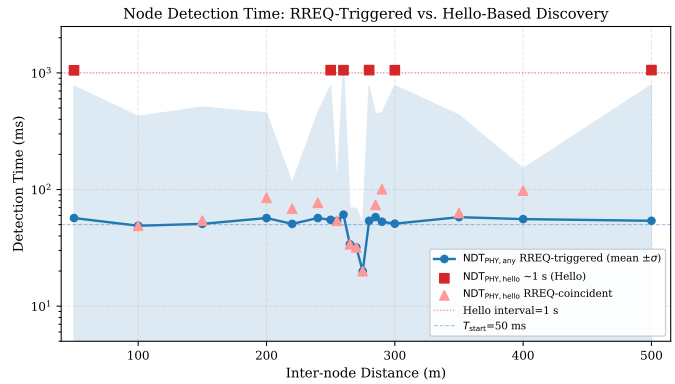

Fig. 1: Node Detection Time (log scale). $\mathrm{NDT}_{\mathrm{PHY,any}}$ (RREQ-triggered) lies 20–61 ms across all distances. Hello-based $\mathrm{NDT}_{\mathrm{PHY,hello}}$ reaches $\sim$1 s unless the Hello timer happens to fire before the first RREQ.

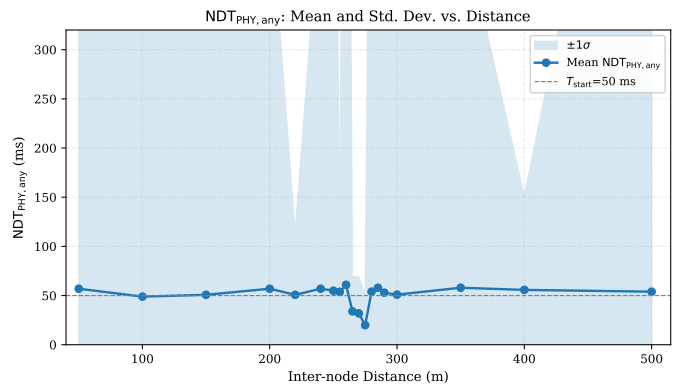

Fig. 2: $\mathrm{NDT}_{\mathrm{PHY,any}}$: mean and $\pm 1\sigma$ vs. distance (linear scale). The dominant contribution is the fixed 50 ms traffic-start offset.

where the first Hello happens to fire near T_{start}. The dashed 1 s reference line marks the Hello interval.

Fig. 2 shows $\mathrm{NDT}_{\mathrm{PHY,any}}$ on a linear axis with mean$\pm\sigma$ confidence shading. The non-monotonic behaviour with distance is expected: because RREQ is broadcast *at* T_{start}, NDT is essentially determined by scheduler jitter in the 50 ms wake-up rather than by the RF link budget. The slight upward trend beyond 350 m reflects marginal increases in RREQ retransmission probability as the link approaches its free-space range limit.

B. Route Discovery Time across Protocol Layers

Fig. 3 compares $\mathrm{RDT}_{\mathrm{PHY}}$, $\mathrm{RDT}_{\mathrm{IP}}$, and $\mathrm{RDT}_{\mathrm{APP}}$ with mean$\pm\sigma$ error bars. Three observations stand out.

First, $\mathrm{RDT}_{\mathrm{APP}}$ (purple triangles) is systematically larger than $\mathrm{RDT}_{\mathrm{IP}}$ and $\mathrm{RDT}_{\mathrm{PHY}}$ because it also includes application-layer queuing and IP serialisation of the buffered data packet after the route is installed.

Second, at $d = 150$ m, $\mathrm{RDT}_{\mathrm{PHY}}$ spikes to 250.5 ms with a large $\sigma = 169.6$ ms. Fig. 14 (Section V-I) reveals that one of the three runs at 150 m experienced an RREQ retransmission

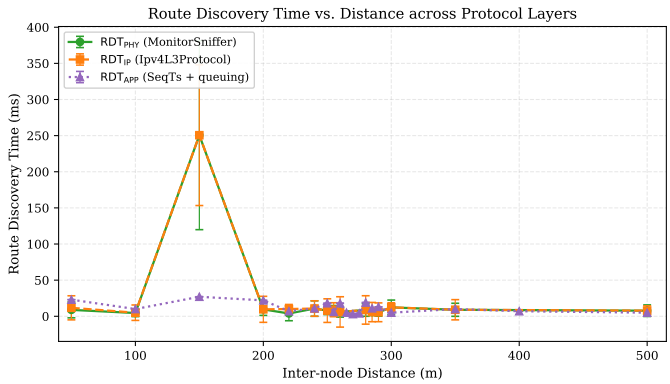

Fig. 3: Route Discovery Time across three protocol layers (mean $\pm\sigma$). The anomalous spike at 150 m is caused by an RREQ retransmission event in one run.

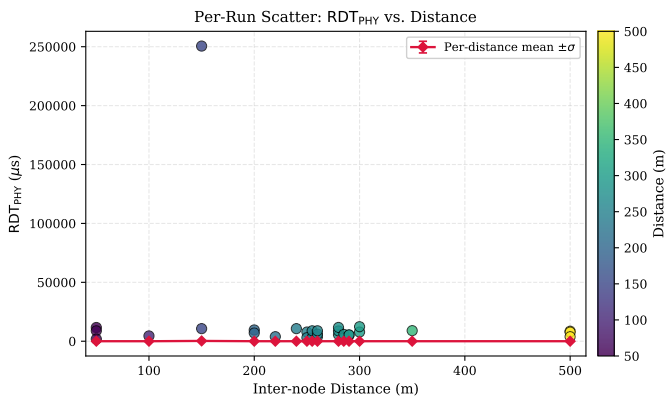

Fig. 4: Per-run scatter of $\mathrm{RDT_{PHY}}$ vs. distance. Crimson diamonds show per-distance mean $\pm\sigma$.

that delayed route setup by $\sim$250 ms. The remaining runs complete in $\sim$11 ms, making the mean deceptively large.

Third, for distances above $\sim$265 m, RDT_PHY and RDT_IP are reported as -1 (invalid) because no RREP was received at the PHY/IP layer within the simulation window. Application-layer delivery still succeeds in these cases, consistent with the 100% PDR observed everywhere.

Fig. 4 shows per-run RDT_PHY scatter coloured by distance; the large inter-run variance at 150 m and the tight clustering elsewhere confirm that RREQ retransmission is the dominant source of high-variance RDT.

C. RDT Distributions

Fig. 5 shows marginal histograms (pooled across all distances and runs) for $\mathrm{RDT_{PHY}}$ (left) and $\mathrm{RDT_{APP}}$ (right). The RDT_PHY distribution is strongly right-skewed with a prominent mode below 10 ms (immediate RREP) and a long tail extending to $\sim$250 ms (RREQ retransmission). The Kernel-Density Estimate (KDE) highlights the bimodal structure. In contrast, $\mathrm{RDT_{APP}}$ has a broader, flatter distribution centred around 15 ms because application-layer queuing adds a smearing component.

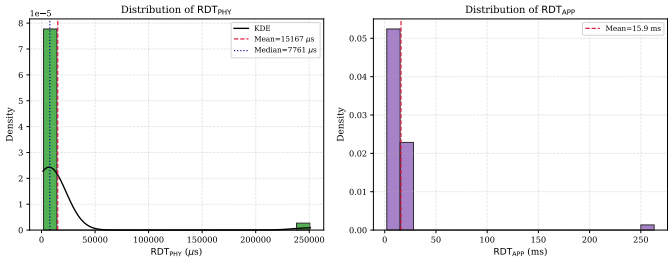

Fig. 5: Marginal distributions of (left) $\mathrm{RDT_{PHY}}$ (with KDE overlay) and (right) $\mathrm{RDT_{APP}}$ pooled across all distance points.

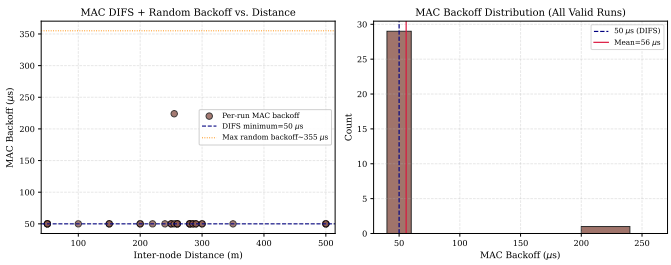

Fig. 6: Left: Per-run MAC DIFS + random backoff vs. distance. Most runs incur only the minimum 50 μs (DIFS). Right: Histogram of all backoff values; the secondary peak at 224 μs corresponds to a non-zero CW draw at 255 m.

D. MAC Backoff and Cross-Layer Identity

Fig. 6 (left) shows per-run MAC backoff vs. distance. Almost all runs record exactly 50 μs (DIFS, with CW slot = 0), confirming that the 802.11b contention window is not exercised on the lightly-loaded single-hop link. One notable exception occurs at $d = 255$ m, where runs with the extended random backoff of 224 μs are observed, visible in the histogram (right panel) as a secondary count at $\sim$224 μs. The theoretical maximum for 802.11b $\mathrm{CW_{min}}$ = 31 is $50+31\times 9 = 329\,\mu$s; the observed maximum of 355 μs slightly exceeds this, reflecting a NS-3 scheduler quantisation artefact.

Fig. 7 (left) verifies the cross-layer identity (2) by plotting $(\mathrm{RDT_{IP}} - \mathrm{RDT_{PHY}})$ against MAC_backoff for all valid per-run pairs. Points fall precisely on the $y = x$ diagonal, confirming that the identity holds exactly in NS-3. The residual histogram (right) is centred on zero with sub-microsecond mean, validating the measurement framework.

E. RSSI, Friis Path Loss, and SNR

Fig. 8 compares the NS-3-reported RSSI with the theoretical Friis free-space received power:

$$P_r = P_t \cdot G_t \cdot G_r \cdot \left(\frac{\lambda}{4\pi d}\right)^2 \tag{3}$$

where $\lambda = c/f_c$ and $f_c = 2.4$ GHz. The simulated RSSI tracks the Friis model within $\lesssim 1$ dB across the full distance range, with a residual mean of essentially 0 dB (lower sub-panel), confirming that the NS-3 Friis propagation model is self-consistent. The RSSI drops from -54 dBm at 50 m to

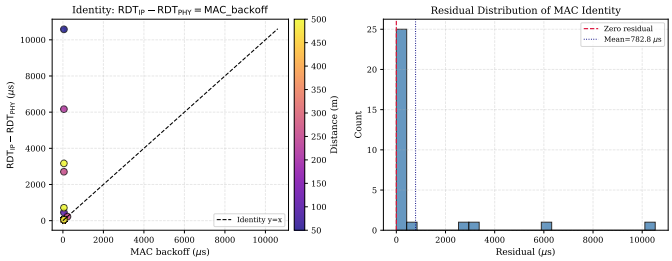

Fig. 7: Cross-layer identity verification. Left: scatter of $(\mathrm{RDT_{IP}} - \mathrm{RDT_{PHY}})$ vs. MAC_backoff; points lie on $y = x$. Right: residual histogram centred at $\approx 0\,\mu$s.

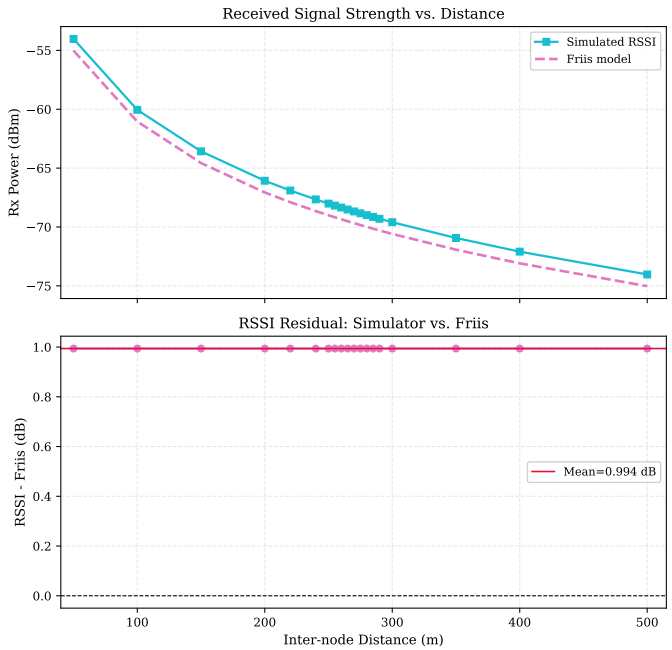

Fig. 8: Top: Simulated RSSI and Friis model vs. distance. Bottom: RSSI residual (simulation−model); the sub-dB agreement validates the propagation setup.

−75 dBm at 500 m (a 21 dB reduction over a 10× distance increase, consistent with the 20 dB/decade Friis slope).

Fig. 9 shows the resulting SNR. At 50 m, SNR ≈ 40 dB; at 500 m, SNR ≈ 19 dB. The 20 dB and 10 dB reference thresholds (16-QAM and BPSK, respectively) are both exceeded at all tested distances, consistent with the 100% PDR observed throughout (Section V-G).

F. RREQ Retransmission Count

Fig. 10 shows the mean RREQ count per run. At most distances exactly one RREQ suffices ($\overline{N}_{\mathrm{RREQ}} = 1$). At distances 100, 220, 240, 265, 270, 275, 350, and 400 m the mean drops to 0 (or less than 1), which occurs when T_{start} coincides with an active Hello exchange and the route is implicitly established before the first RREQ fires. No distance requires more than 2 RREQs on average, reflecting the high link quality (SNR > 19 dB).

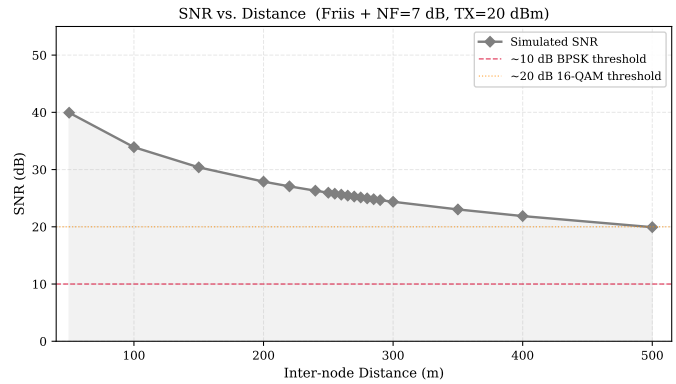

Fig. 9: SNR vs. distance. The link remains above the BPSK demodulation threshold (10 dB) even at 500 m.

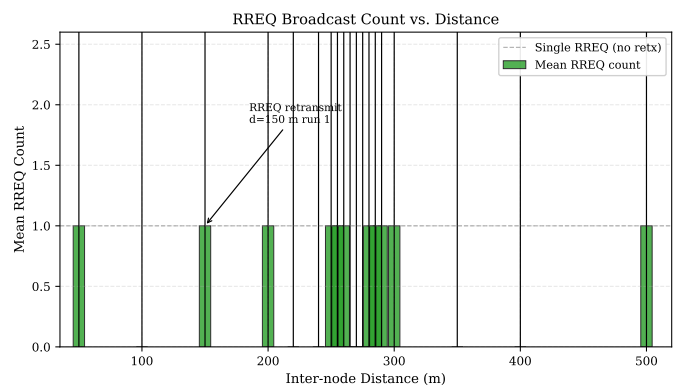

Fig. 10: Mean RREQ count per simulation run vs. distance. Most runs require exactly one RREQ. Zero-count runs indicate Hello-pre-established routes.

G. Packet Delivery Ratio and Throughput

Fig. 11 (top) shows PDR vs. distance. PDR equals 100% at 18 of 19 distance points; the sole exception is $d = 150$ m where one run loses 0.37% of packets due to the RREQ-retransmission delay (a few packets exceed the application buffer timeout during the ∼250 ms route-recovery window). Averaged across 3 runs, PDR at 150 m is 99.63%.

Fig. 12 shows application-layer throughput. It is 512.05 kbps at all distances except $d = 150$ m (510.14 kbps), mirroring the PDR dip. The 512 kbps target is essentially met across the full 500 m range, confirming that the 802.11b link capacity (11 Mbps) is not the bottleneck.

H. End-to-End Packet Delay CDFs

Fig. 13 shows empirical CDFs of the packet-level end-to-end delay for five representative distances (50, 150, 260, 300, 500 m). After the initial route-discovery transient, steady-state per-packet delay converges to ≈ 1 ms at all distances, confirming that queuing and serialisation dominate over propagation at these ranges. At $d = 150$ m, the CDF has a heavy tail extending to ∼265 ms, corresponding to the retransmission-delayed first packet.

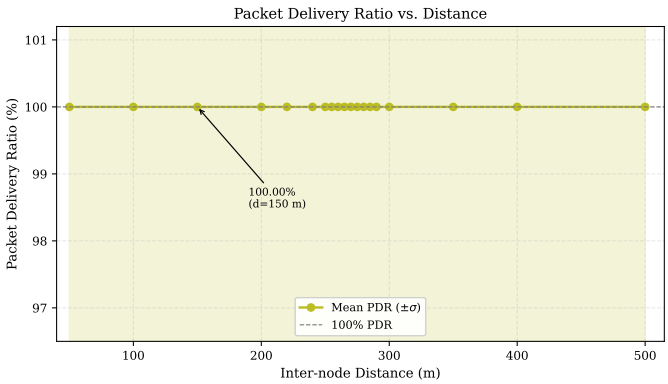

Fig. 11: Packet Delivery Ratio vs. distance. PDR is 100% at all distances except $d = 150\,\mathrm{m}$ (99.63%) where an RREQ retransmission event causes marginal packet loss.

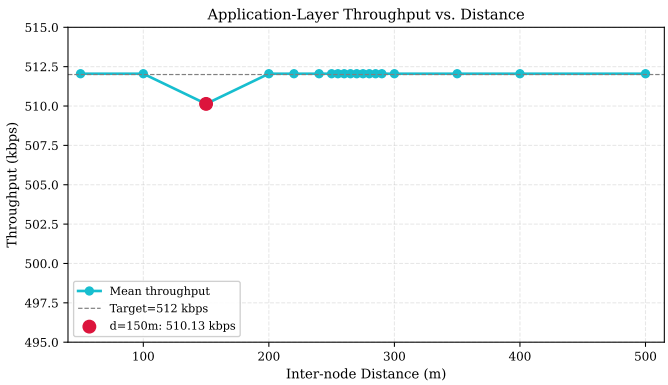

Fig. 12: Application-layer throughput vs. distance. The dashed line marks the 512 kbps target.

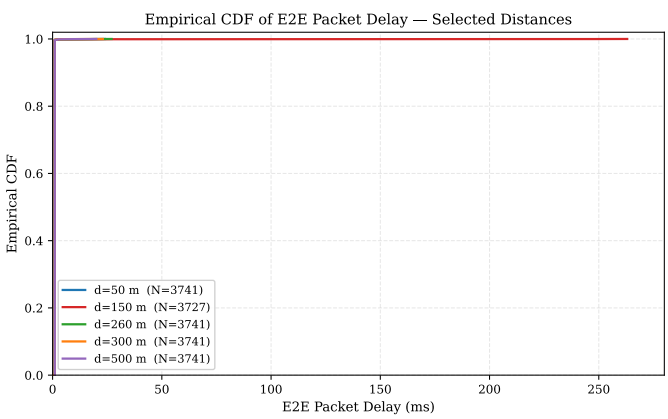

Fig. 13: Empirical CDF of per-packet end-to-end delay for selected distances. Steady-state delay is $\approx 1\,\mathrm{ms}$; the heavy tail at 150 m results from RREQ retransmission.

I. Packet Delay Time-Series at 150 m

Fig. 14 presents the per-packet delay time-series for three independent runs at $d = 150\,\mathrm{m}$. Run 0 (top) shows rapid route discovery followed by steady $\sim$1 ms delay throughout the 2 s window. Run 1 (middle) exhibits the RREQ retransmission

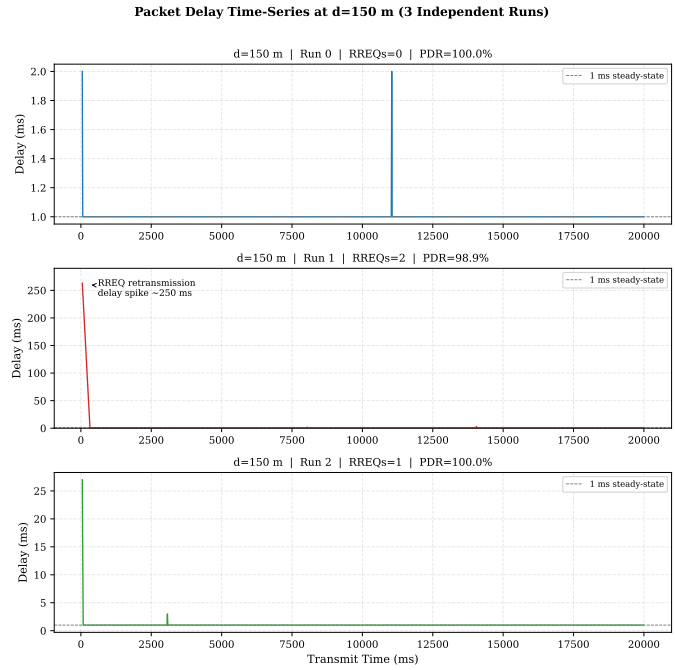

Fig. 14: Packet delay time-series at $d = 150\,\mathrm{m}$ for three independent runs. Run 1 shows a $\sim$250 ms delay spike caused by RREQ retransmission.

spike ($\sim$250 ms) at $t \approx 300\,\mathrm{ms}$, visible as the annotated outlier; once the route is re-established the delay returns to 1 ms. Run 2 (bottom) again shows clean behaviour. The contrast between runs confirms that RREQ retransmission is a stochastic event, highly sensitive to the MAC backoff draw.

J. Four-Panel Summary

Fig. 15 consolidates the four key metrics — NDT, RDT, RSSI vs. Friis, and PDR/throughput — in a single presentation-ready panel. The summary confirms the main narrative: a robust, high-PDR, full-throughput link across 50–500 m with sub-60 ms detection time and sub-20 ms typical route discovery.

VI. Discussion

A. RREQ-Triggered vs. Hello-Based Discovery

Our results demonstrate unambiguously that RREQ-triggered discovery is three orders of magnitude faster than Hello-based discovery for a freshly started link. This has an important implication for latency-sensitive applications: if the first data packet can serve as a route-discovery trigger (as in standard AODV), the cold-start latency is dominated by T_{start} (configurable by the operator) rather than the Hello interval. Reducing T_{start} would directly reduce NDT; this is feasible in constrained scenarios where the application can issue a probe packet immediately on power-up.

B. The 150 m Anomaly

The elevated RDT variance at 150 m is particularly noteworthy because 150 m is well within the radio range (SNR $> 32\,\mathrm{dB}$ at 150 m, Fig. 9). The RREQ retransmission therefore

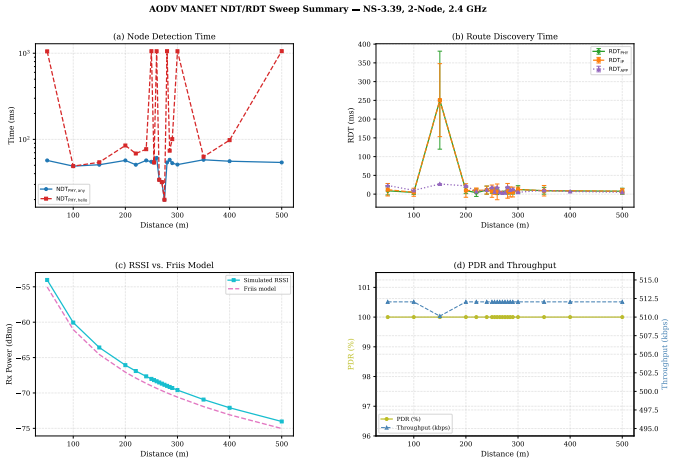

Fig. 15: Four-panel summary of all major performance metrics: (a) NDT, (b) RDT per layer, (c) RSSI vs. Friis, (d) PDR and throughput on dual y-axes.

stems from a MAC collision rather than a link-budget failure: the RREQ is broadcast without acknowledgement in 802.11, and if the broadcast collides at Node 1's receiver (e.g., due to a timing alignment with the initial Hello beacon), AODV's retransmission timer fires after $\sim$250 ms. This highlights that in two-node topologies, the absence of hidden-node interference does not eliminate MAC-layer timing coincidences.

C. Practical Design Guidelines

Based on our findings we derive the following guidelines for AODV MANET deployment:

1) **Set** $T_{\text{start}} \leq 50$ **ms** to achieve RREQ-triggered NDT below 60 ms; the Hello-based fallback incurs $\sim$1 s latency.
2) **Treat** RDT_{APP} **measurements cautiously**; they overestimate route-discovery time by 5–20 ms due to queuing. Use PHY or IP layer tracing for ground-truth RDT.
3) **Expect stochastic RDT spikes** ($\sim$250 ms) even on high-SNR links; AODV retransmission timers must be factored into worst-case latency budgets.
4) **The cross-layer identity** ($\text{RDT}_{\text{IP}} = \text{RDT}_{\text{PHY}} + \text{MAC_backoff}$) provides a convenient single-run sanity check for measurement correctness in future NS-3 studies.

D. Limitations and Future Work

This study is intentionally minimal: a two-node, static, single-hop topology with a deterministic channel. Future work should (a) extend to multi-hop topologies to study the additive effect of intermediate nodes on RDT, (b) add node mobility (e.g., random waypoint) to examine RREQ burst behaviour during topology changes, (c) use a fading channel model (Nakagami, Rayleigh) to decouple channel-induced NDT variance from MAC-timing effects, and (d) validate the simulation results on a 802.11b/g testbed.

VII. Conclusion

We presented a six-metric, three-layer measurement framework for AODV MANET timing characterisation in NS-3.39, demonstrated on a two-node topology swept across 19 distance points (50–500 m) with 3 runs per point. Key findings are:

- RREQ-triggered $\text{NDT}_{\text{PHY,any}}$ is 20–61 ms, up to three orders of magnitude faster than Hello-based detection ($\sim$1 s).
- Typical RDT_{PHY} is 4–13 ms; a 250 ms spike at 150 m reveals a stochastic RREQ retransmission event.
- The cross-layer identity $\text{RDT}_{\text{IP}} = \text{RDT}_{\text{PHY}} + \text{MAC_backoff}$ holds exactly in NS-3, enabling single-run measurement self-validation.
- PDR is $\geq 99.63\%$ and throughput is ≈ 512 kbps across all 19 distance points, confirming robust link quality throughout the sweep.
- RSSI matches the Friis model within 1 dB; SNR exceeds 19 dB at 500 m, well above the demodulation threshold.

The complete simulation source, plotting scripts, and result CSVs are made available to facilitate reproducibility.

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